Beyond Score Prediction:
Evidence-First Reasoning for Writing Assessment

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Abstract

Gnomon constructs writing-assessment scores from rubric-linked observations of the learner's text. A compiled rubric supplies both extraction and scoring rules; verification checks source quotations before they earn credit. We describe exact-span recovery and relation-level duplicate-credit control, candidate-bound correction selection, and completion-weighted training with chunked vocabulary projection. The accompanying data pipeline covers source selection, task and image admission, and training preparation. An archived-input replay demonstrates how verified evidence produces a criterion score. Historical Task-2 development records report MAEs of 0.226–0.319 on 157 essays under source-specific calibration. This systems report explains the implementation and its measurement conditions, separating observed results from the comparisons needed to assess generalization.

1 Introduction

A writing score is easier to review when its supporting observations and scoring rules can be inspected together. Gnomon records those links so a reviewer can trace a disputed judgement from the learner's text to the criterion score.

Gnomon separates *observation*, *verification* and *scoring authority*. A model extracts criterion-level evidence; tool and verification components assess it; explicit rules construct criterion scores. Here, *reasoning* means inspectable transformations, not access to a model's hidden cognition.

IELTS Writing uses four criteria per task: Task Achievement (TA) for Task 1 or Task Response (TR) for Task 2, plus Coherence and Cohesion (CC), Lexical Resource (LR), and Grammatical Range and Accuracy (GRA). Five criterion names across tasks do not imply five scores on every essay. General Training Task 1 uses a letter instruction rather than an Academic chart (IELTS, 2023a,b).

Contributions. We describe three parts of the implementation: (1) a shared rubric for extraction and scoring, with exact-span verification and relation-level credit accounting; (2) a source-aware learning loop that binds corrections to the attempts they revise; and (3) completion-focused training that preserves example weights across microbatches and records supervised-token throughput. An archived-input replay shows the scoring process; population records distinguish corpus inventory, prepared examples and a historical training run. The research contribution is how these components work together.

2 Related work

Interpretable, feature-informed scoring and criterion-specific assessment are established research directions. e-rater V.2 and the broader automated-essay-scoring literature provide the context for this work (Attali and Burstein, 2006; Ke and Ng, 2019). Gnomon studies how data governance, rubric definitions, evidence transport and constrained scoring can be connected through inspectable interfaces.

Automated-scoring evaluation must be tied to intended use, with agreement, error, bias and validity evidence considered together (Williamson et al., 2012). Structured model judging also remains vulnerable to position, verbosity and self-enhancement biases; an inspectable format does not eliminate them (Zheng et al., 2023; Liu et al., 2023).

3 System architecture

Figure 1 shows the connected information flow. The inspected *grade_engine* entry calls *eval_harness*, which assembles evidence for

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scorer_3layer. Rubric content supplies both elicitation and interpretation.

Descriptions distinguish the *default* scoring route, *conditional* branches enabled by configuration, *historical* runs and *prospective* studies. This prevents a component implemented in one configuration from being attributed to every run.

3.1 Declarative rubric and extraction contracts

The rubric specification defines constructs, items, dimensions, graded groups, task variations and evidence requirements. The *compile_rubric* implementation validates it, builds the runtime item bank and assigns a content-based version. Prompt construction and scoring read the same item definitions. Registered dependencies can be updated to the new version; optional additions remain separate from the default bank.

Task metadata selects the extraction route. Its input includes text and, where applicable, an image; the request builder supports task instructions and criterion-specific evidence schemas. The model returns locatable quotations associated with rubric items rather than an essay-band prediction. Complete task-context propagation must be checked for each configured route.

The extraction contract restricts model output to observations, while scoring rules and calibration references reside in their own components. This is an information boundary, not a claim that all upstream selection is label-free.

3.2 Verification and constrained judgement

The shared scoring consumer combines model-derived item verdicts with deterministic tool signals. It retains distinctions among supported, partially supported, unsupported and unmeasured evidence. Unmeasured evidence and failed presence checks are distinct. In R1, rejection of the legacy flow quotation leaves its presence-rule item UNMET, while the chain and pivot channels return NO_EVIDENCE and are excluded from dimension combination. Zero admissible facts therefore does not automatically mean an unmeasured item.

The default scoring path maps item evidence to criterion dimensions, accounts for graded-ladder structure and applies limiting-dimension aggregation with non-compensatory constraints. A dimension at the top of its measured ladder supplies a lower bound rather than automatically limiting a deeper dimension. Configured psychometric or adopted magnitude-estimation paths can change the base estimate; neither is a universal default. The Rasch branch requires *SCORER_MATH=rasch* and calls *rasch_band* with credited evidence. The complete bank and calibration recipes are not distributed.

For a populated criterion, the order is dimension credit and ladder accounting, limiting aggregation, optional base-estimate substitution, corroborated floor/cap application, half-band rounding, then configured low-tier scale and high-tier uncertainty hooks. The evaluation consumer applies deterministic guardrails after scoring; the engine's decision and human-release boundaries remain downstream. The inspected scale and high-tier hooks are disabled/identity by configuration. An empty dimension set takes the earlier fallback in Table 1, rather than traversing the populated combine.

The retained trace lets a reviewer ask separately whether a quotation exists, whether its interpretation is sound, and which scoring condition it affected.

Decision lineage. A review separates *what was observed*, *what earned credit*, and *what was released*. The response and task identify the assessment; the rubric version fixes the item definitions; verified observations feed the scorer; and human review governs learner-facing release. When a reviewer rejects an observation, its affected scoring rule can be located without treating the revised score as new evidence. Shared definitions reduce mismatches between prompting and scoring, while route-level checks establish which links a particular execution preserved.

3.3 Abstention, cross-checks and correction

Disagreement checks can withhold a score when estimates conflict. A single estimate provides no agreement-based confidence evidence, and correlated repeated calls are not independent raters. Evaluation must therefore report coverage alongside error among accepted cases.

The wider engine also contains judge and correction branches. These can cross-check signals, regenerate flagged evidence and rescore it through the shared consumer. Production automatic checks are deterministic-only; model-backed correction and the provenance-bound teaching overlay require explicit modes, callbacks and context. The selected mode determines which branch a report tra-

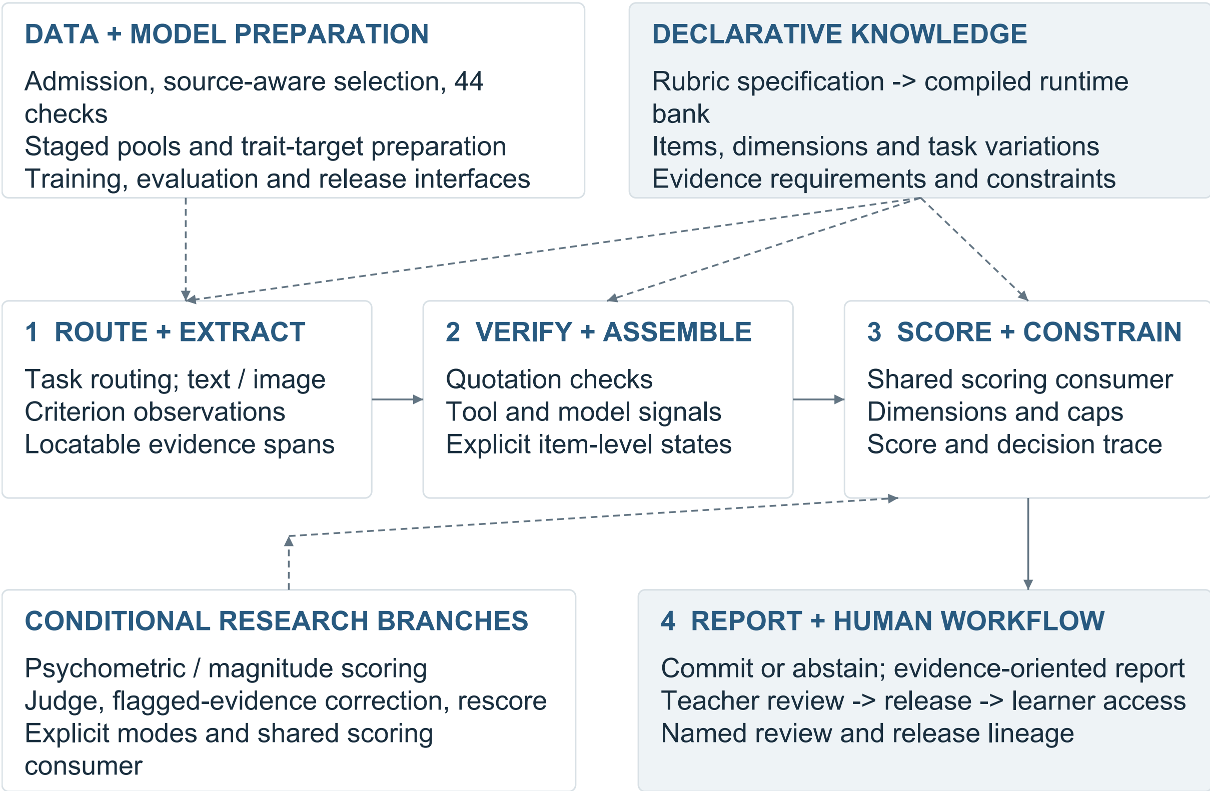

Figure 1: The central assessment path and supporting systems. Dashed links identify configuration or conditional branches; human release remains a separate authority step.

verses.

The delivery workflow supports teacher review, release, learner access and challenge. Learner-facing feedback must translate a verification reason into a clear diagnosis and manageable action. Its helpfulness requires evaluation with teachers and learners.

3.4 Exact-span transport and relation identity

Supported extraction items can use sentence locators and short head/tail anchors instead of reproducing a long quotation. The resolver checks the permitted structure, contiguous sentence range, unique matches and ordering, then slices the original response. It does not repair spelling or rewrite the learner's text. Invalid transport rejects the emission or omits the affected fact with an issue, according to the caller's mode.

The relation verifier works above individual quotations. It checks endpoint grounding, cardinality, permitted item memberships, resolution state, scope and agreement among repeated views. Conflicting identities or inconsistent copies invalidate the relation. An essay-local index retains valid views while assigning a resolved relation's positive credit only to its first eligible item in the immutable rubric order. Repeated presentation of the same relation therefore does not create repeated credit. This is mechanical identity accounting, not a judgement that the relation is semantically correct.

These checks separate quotation recovery, relation identity and interpretation. A reviewer can locate whether an error arose in extraction, verification or scoring; correct quotation recovery alone does not establish a correct interpretation.

3.5 Worked extraction-to-score replay

Trace R1 executes the current shared verification, tool, derived-item, scoring and guardrail consumers on the first existing CC cache record, selected before inspecting its result. The archived response and extraction are bound to file, row and text hashes; the replay records the runtime-bank and owner hashes. It uses the heuristic configuration with no model call, new extraction or provider access.

Table 2 presents selected trace observations, not

Evidence state	Implemented consequence
Supported / unsupported	MET and UNMET are measured states. Credit follows the item's role; a measured flaw can activate a configured cap. An unsupported positive claim is not automatically a flaw.
Partially supported	Graduated items can receive intermediate credit. Neutral and prerequisite items do not receive positive partial credit; unmet measured prerequisites can prevent a higher rung from earning credit.
Unmeasured / unavailable	NO_EVIDENCE, omitted tool signals and unmeasured items do not testify as measured failures. A dimension with no measured item is excluded from combination.
No measured dimensions	The scoring component returns its configured floor, then applies its configured scale/uncertainty hooks. In the inspected CC configuration the result is 4.0, not an automatic abstention. This is distinct from having no score estimate.
Missing task context	Task-scoped checks can be skipped when identity is unavailable. A grounded quotation alone cannot certify task fulfilment. Task-aware admission and upstream input controls are separate contracts.
Decision without estimates	The disagreement layer abstains when its estimate list is empty. With one estimate it commits that estimate; this supplies no independent agreement evidence.

Table 1: Evidence states are not interchangeable. These are source-bound component semantics, not a claim that every route enforces a universal missing-context refusal.

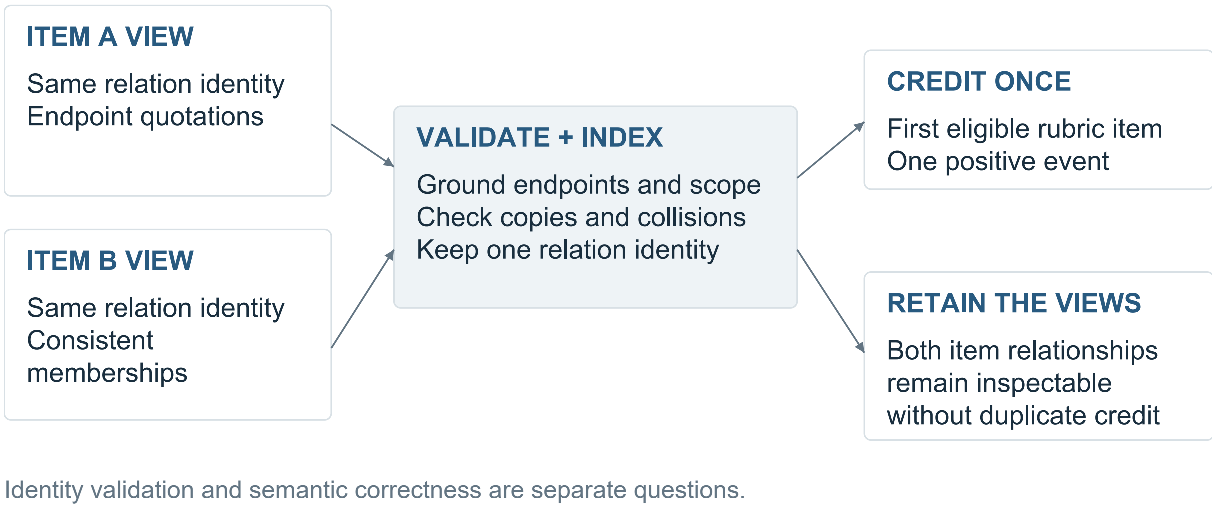

Figure 2: Schematic relation-level accounting. Several item views can refer to one validated relation; scoring credits it once. No learner text or private rubric item is shown.

an exhaustive scoring derivation. The dimension positions and logged base of 5.30 also depend on other verified facts in the archived row; the linker examples alone do not determine 5.5. The complete essay and private bank weights are not supplied for independent numerical reproduction.

The original cached input is unchanged. As a separate negative control, replacing the only retained entity-carry-over quotation with absent text changes that item's verdict from MET to UNMET. This tests the grounding boundary, not whether the original entity interpretation was correct. The archived row has no task identity, so task-scoped completeness is not tested; R1 is a compatibility replay, not a claim of current extraction quality or a full learner workflow. Its 5.5 is one computed criterion score, not an MAE.

4 Data and learning infrastructure

4.1 Quality controls with explicit scope

The *run_full_qa_gate* registry contains 44 distinct checks across seven groups. Figure 3 shows that implemented surface. It describes coverage, not a current pass rate.

Controls address exact and normalized duplication, near-duplicate detection, cross-split overlap, missing or malformed fields, source-label trust, distribution coverage and training-readiness con-

Boundary	Observed R1 trace
Source and proposal	The response contains “However” and “On the other hand”. The cached model proposes these as linker observations; the discarded “By the way” proposal is not positive evidence.
Grounding and rule	Two retained linker quotations satisfy the configured presence rule: the attempted-cohesion item is MET. A concatenated, noncontiguous quotation proposed for an overlinking flaw contributes zero grounded facts; that flaw rule is UNMET.
Transport rejection	A legacy raw quotation targets a now anchor-only flow item. The resolver records ANCHOR DROP; zero admissible facts leave this presence-rule item UNMET. Relation identifiers are absent in this legacy record; relation-level duplicate credit is not exercised.
Dimension consequence	The resulting coherence position is 5.0; cohesion is 7.0. Cohesion reaches its measured ladder top and becomes a lower-bound observation, while coherence remains limiting.
Score construction	The limiting-weighted base is 5.30. No cap or guardrail changes this case; half-band rounding produces 5.5. Passing this single estimate to the decision component returns COMMIT. This is not a human-approved release.

Table 2: Selected observations from an executed archived-input replay, not an exhaustive scoring derivation or accuracy test. The trace does not establish that the model’s semantic interpretations are correct.

ditions. The implementation distinguishes reporting, review requirements and selected repair actions. The training sub-gate selects eight of the registered checks; it is not equivalent to running the entire quality surface.

A central selection API filters by source, task, role, quarantine and evaluation exclusions, then stratifies the selected pools. Each discovered store requires an assigned role and either a named consumer or an explicit deferral. Some orchestration links are declared but not yet executed.

Practice-pool preparation stages all five criterion outputs, validates them with their reader, preserves previous versions and publishes the manifest last. Atomic replacement applies to each file, not to the set as a whole.

4.2 Documented asset and run scale

The September inspection reconciled the inventory, parsed the two prepared SFT files and inspected a separate historical training manifest. The counts below have different units and overlap; they are not an additive evaluation population.

The current prepared files and the earlier checkpoint have separate provenance.

4.3 Admission and supervision construction

Training admission checks identity, usable text, task/criterion compatibility, held-out membership, quarantine and selected quality flags. Academic Task-1 achievement examples require a local image through the shared validity owner; specified score-label patterns are rejected from completions. Named reasons make exclusions countable. Label-agnostic teaching can skip label-distribution flags while retaining text-quality and integrity checks. Retired per-essay AI-detection scores are not active authorship gates.

A contrastive assembler uses source-score quantiles to select high/low pairs, then emits paired texts with criterion and dimension identifiers but without numeric scores. Selection is label-dependent; only the emitted representation omits absolute scores. A separate weighted absolute-target assembly is maintained as an optional ablation. Supervision format is therefore an inspectable choice rather than an implicit mixture.

Governed library-write tooling records authorisation, snapshots and before/after row counts alongside the mutation. Together with staged pool publication, it provides a lineage record from selected inputs to emitted artifacts.

4.4 Psychometric target preparation

Three measurement concerns are treated separately: targets used during preparation, evidence-to-score calibration, and confidence monitoring.

Trait-target calibration is separate from the contrastive assembler in Section 4.3. Despite its historical MFRM filename, the preparer fits per-trait PCM targets or uses prompt-centred normalization. The latter is a fallback, not a fitted item-response model. Joint rater-severity estimation remains a proposed extension.

Parameter-efficient multimodal adaptation uses established LoRA and QLoRA methods (Hu et al., 2022; Dettmers et al., 2023), with completion-focused supervision. The private curriculum, prompts, adapter settings and calibration artefacts

Figure 3: Static registration counts independently reconciled against the source list. Groups are organizational categories, not statistically independent controls or observed quality improvements.

Record	Recorded scope	Interpretation
Corpus inventory	101,175 records; 47 source labels	Generated inventory across heterogeneous sources.
Prepared SFT data	10,888 train rows; 578 validation rows	11,466 parsed preparation records.
Earlier 8B run	9,325 examples; 2,134 image-bearing	Separate non-smoke run; curriculum and weighting enabled.

Table 3: Distinct data-engineering and training records. Source labels are categories, not a count of independent corpora. Preparation rows and historical examples are separate populations.

are not distributed.

4.5 Critique and repair selection

The *raft_select* module groups model attempts by essay and ranks eligible attempts by a faithfulness reward. Auxiliary judgement diagnostics are recorded separately and do not select the training target.

For an essay with k passing attempts out of n, the selector keeps a best passing attempt from the mixed region $0 < k < n$. All-pass examples are skipped by this rule; all-fail examples become teacher-rescue candidates. The emitted completion is serialised from the scored extraction, so a caller cannot substitute a different training completion after selection.

Pass predicate and reward. The evaluated object is the structured extraction and its source response. The *verifier_reward* owner requires quote grounding, at least 90% valid fact schemas, no forbidden score-label leakage, sufficient factual coverage for a long response, and passing anti-copy and distinct-evidence checks. When submission-identity arguments are supplied, task/image identity is checked first; legacy callers supplying none skip that contract. A hard-gate failure cannot be rescued by a higher reward.

The ranking reward combines grounding, schema validity, absence of forbidden score labels and evidence coverage, with a small bonus when an attempt includes both retained and discarded observations. Copy detection and distinct-evidence checks determine eligibility only. These tests reject specified extraction failures; semantic accuracy and learner usefulness require separate evaluation.

The critique path binds the original candidate, issue set and repair. A revision must differ from the attempt, satisfy the same gate and improve the

Mechanism	Engineering role	Qualification
Trait-target preparation	Per-trait partial-credit-model estimation where available; prompt-centred normalization fallback.	PCM when fitting is available; normalization otherwise.
Ranking-aware supervision	Continuous trait-target loss combined with adjacent-pair ranking constraints.	Implemented training-loss interface.
Evidence calibration	Conditional Rasch and graded-response modelling, uncertainty estimates and item-fit diagnostics.	Gated research path, not the default scorer.
Scale and source checks	Monotonic score-scale mapping and cross-source item-invariance diagnostics.	Not joint rater-severity estimation.
Confidence monitoring	Prospective decision/outcome joins and confidence-bin summaries.	Separate from grading-accuracy validation.

Table 4: Implemented calibration-related mechanisms and their roles.

base reward:

$$R(e_{\text{revision}}) - R(e_{\text{attempt}}) > 0.$$

STOP responses, invalid bindings, failed revisions and non-positive reward changes are recorded rather than admitted. Each correction is keyed by criterion and mechanism/item identifier, not free-text similarity. Matching keys from first-pass completions serve as anchors; a correction without an anchor is held for review. This preserves first-pass supervision alongside repair examples. Increasing the verifier reward alone does not demonstrate better teaching.

Training preparation mixes base data, selected attempts and verified corrections while controlling the weight of self-generated examples. The resulting package supports verifier-filtered SFT. Candidate evaluation and adoption follow training; separate policy-optimisation components are not evidence of a completed RLVR run.

5 Training systems and resource accounting

5.1 Completion-focused adaptation

The portable trainer resolves model and precision configuration, checks trainable parameter scope and constructs multimodal batches. Completion-focused supervision masks non-target tokens; per-example weights are carried into the loss interface. Curriculum ordering and length grouping are explicit alternatives: phase order versus grouping supplied sequence lengths to reduce padding. The implementation rejects their simultaneous selection rather than claiming both ordering properties.

5.2 Chunked vocabulary projection

The weighted loss backend projects completion positions through a frozen language-model head in vocabulary chunks. Streaming softmax statistics support full-vocabulary cross-entropy and an entropy term without materialising every sequence position's full logits at once. Backward computation accumulates hidden-state gradients over the chunks; the head must remain frozen.

For a logical optimiser window, the implemented objective is

$$\mathcal{L} = \mathrm{WM}_w(\overline{\mathrm{CE}}) - \beta\, \overline{H}_{\text{completion}},$$

where WM_w is the weighted mean across examples, normalised by the sum of their weights; $\overline{\mathrm{CE}}_i$ averages the completion-token cross-entropy of example i. The entropy mean covers all completion tokens in the window. Common window denominators preserve the intended weighting across microbatches. A materialised reference supplies an arithmetic comparison target. Floating-point reduction order can differ; no new kernel-performance measurement is reported.

Averaging microbatch means would give a small microbatch the same influence as a large one. Logical-window denominators instead preserve example weights for cross-entropy and token weights for entropy. Vocabulary chunking controls when logits are materialised; it does not change the intended weighting.

5.3 Useful-work denominators

The trainer separately records input tokens, non-padding tokens and completion-loss tokens from windows whose optimiser steps actually applied. These describe batch throughput, useful-token throughput and supervised-target throughput. Keeping them distinct prevents padding from being credited as the same unit of learning work.

Admission, checkpoint evaluation, packaging and serving interfaces connect training to release decisions. Training completion, model selection

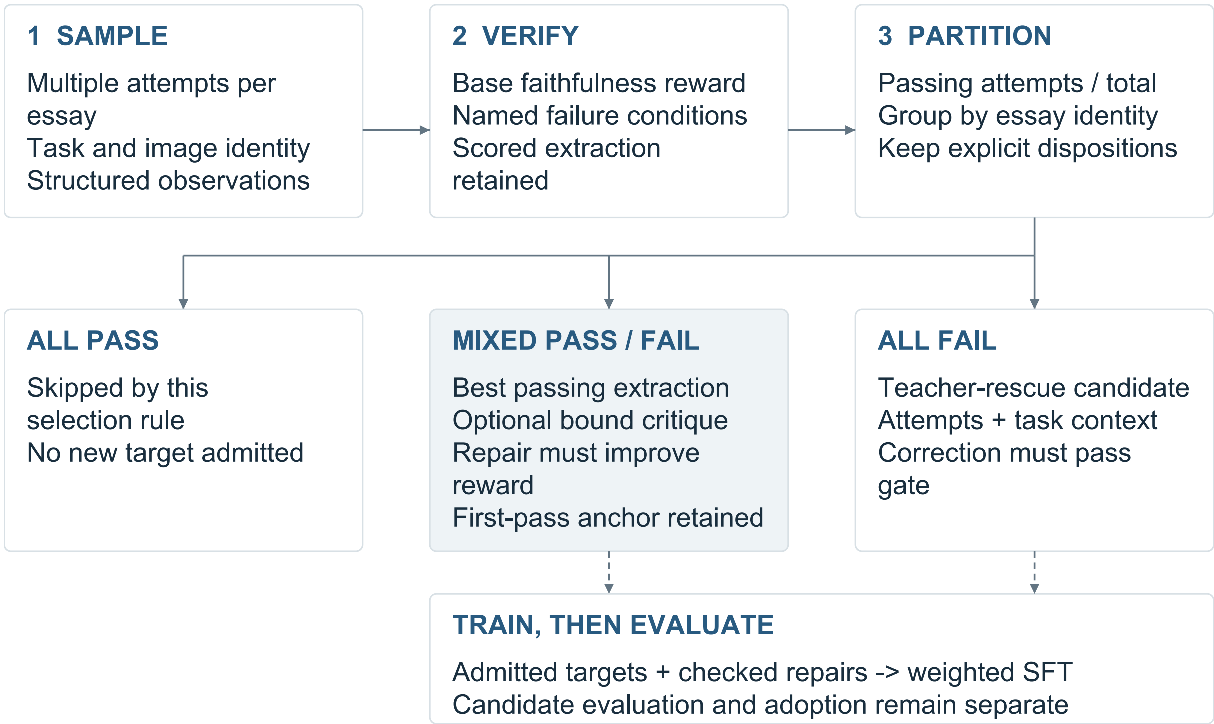

Figure 4: Implemented selection and training interfaces. This is a configured research loop, not a record of a completed production continual-learning campaign. Passing refers to the specified verifier conditions.

and serving admission remain separate states. The historical 8B manifest records one execution, not a claim that later trainer components ran in that experiment.

5.4 Earlier encoder engineering

Historical ModernBERT work includes per-layer low-rank adapters, fused/separate projection handling and inference-time merging. A DirectML-compatible AdamW path preserves optimiser state and bias correction using supported updates. These are earlier implementations, not dependencies of the current extractor.

6 Evaluation scope and observations

6.1 Study method

This study examines the assessment pipeline's implementation, data records and recorded results. It combines source inspection, reconciliation of population counts, an archived-input scoring replay and reproduction of the aggregate figures.

The September 2026 review followed the project registries to 46 source files, then traced admission, relation handling, correction selection and training loss through their callers. A private evidence index links each implementation claim to the inspected source version. Data and run counts were checked against their respective records.

The accompanying *build_figures.py* defines three implementation schematics and two aggregate plots. Its numerical inputs are also exported as *figure_data.json*. Running *python build_figures.py* with ReportLab and Arial/Arial Bold redraws those figures and checks the aggregate arithmetic; the schematic definitions are not measurements. This reproduces the displayed figures, not the original assessment experiment. R1's separate replay script invokes the existing scoring owners on its pinned archived input and emits the resulting trace and checks. It requires the private input and matching runtime, unlike the aggregate figure generator.

6.2 Historical grading results

The earlier Task-2 system reported criterion MAEs of TR 0.319, CC 0.268, LR 0.248 and GRA 0.226 on 157 essays under source-specific calibration. Recorded score correlations ranged from 0.915 to 0.941. Figure 5 presents these development results.

These aggregates describe the earlier system and its recorded calibration conditions. The complete prediction/reference rows, calibration membership and model configuration have not been re-

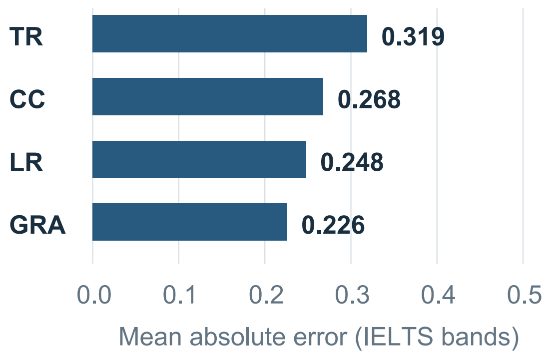

Figure 5: Historical Task-2 MAE on 157 essays with source-specific calibration. Reported correlations: TR 0.941, CC 0.915, LR 0.919 and GRA 0.921. These aggregates are not a current-engine or independently reproduced held-out benchmark.

constructed here, so the report does not estimate held-out generalization or a matched advantage over another grader.

7 Discussion

Gnomon makes the score reviewable at several levels: source quotation, rubric interpretation, evidence credit and scoring rule. A teacher or developer can challenge a specific step instead of accepting or rejecting the entire model response. Data selection and correction training use corresponding identifiers, linking assessment review to subsequent learning targets.

The next evaluation should test this design against a direct call to the same model on previously unused, input-complete essays. References and resource budgets should be fixed, with calibration outside the test cohort. Reporting predictions, refusals and exclusions permits comparison of error, coverage and cost. Blinded review of complete feedback, learner comprehension and delayed transfer test teaching outcomes that score agreement alone cannot measure.

8 Conclusion

Gnomon links a compiled rubric, source-grounded observations and explicit scoring rules, with corresponding controls for data preparation and correction training. The report details three reusable mechanisms: exact-span and relation-level credit accounting, candidate-bound repair selection, and completion-weighted training across microbatches. The scoring replay and historical results illustrate the implementation and its development record. Matched evaluations will determine where this design improves grading, feedback and delivery cost.

Limitations

The study is retrospective and configuration-specific. It does not include a matched direct-call benchmark, a teacher-preference study or a learner-transfer experiment. The historical scoring result cannot be transferred to the current engine without re-evaluation. Likewise, registered quality checks describe the data pipeline's controls, not the current dataset's pass rate.

Private datasets, rubric contents, prompts and calibration recipes are not distributed. The aggregate data and figure code reproduce the plots; reproducing an assessment run also requires its inputs, row-level predictions and exact configuration.

Ethical considerations and AI assistance

The intended application is research into formative writing support, not autonomous high-stakes grading or certification. Misinterpreted evidence and unhelpful feedback can mislead learners; source, task and language-background imbalance can produce unequal errors. Human review is a proposed safeguard, not proof those risks have been removed.

The report presents aggregates and short, non-identifying phrases from R1, without distributing the learner response, source record or institutional identifiers. Source licences, consent conditions, examiner qualifications and ethics approvals are outside this inspection and must be established for the intended data sharing, deployment or human study. No official IELTS endorsement is claimed.

Claude and OpenAI Codex assisted implementation, drafting, reference checking and figure-generation code. Three figures are programmatic implementation schematics; two plot disclosed aggregates. AI consultation and tool checks are not independent human peer review. The author retains responsibility for final claims, citations and permitted data use.

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